SIMATS ENGINEERING

ASSESSMENT - IV

COURSE CODE: CSA1509

COURSE NAME: CLOUD COMPUTING & BIG DATA ANALYTICS

Code of Conduct:

I, **Ashira Bridget Susan S (192524316),** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



CO4-Assessment Tasks

1. Define Big Data and explain why traditional data processing approaches are often insufficient for big data environments.

2. Explain the 5Vs of Big Data with one practical example for any two Vs.

3. Differentiate structured, semi-structured, and unstructured data with examples.

4. Explain the role of distributed file systems in big data processing.

5. List any three major security concerns in big data platforms and briefly explain each.

6. What are data appliances and integration tools? Explain their role in big data workflows.

7. Discuss two major challenges organizations face while implementing big data analytics.

8. Write short notes on any one domain application of big data analytics.

9. How do business drivers influence the adoption of big data solutions?

10. Explain the importance of big data analytics in decision making.

CO4 Big Data Concepts: Short and Descriptive Answers

Q1. Define Big Data and explain why traditional data processing approaches are insufficient

Definition

Big Data refers to datasets so large, fast-growing and varied that conventional databases and single-machine tools cannot capture, store, manage or analyse them within a reasonable time and cost. It is commonly characterised by volume, velocity and variety (the 3Vs, later extended to 5Vs), and it demands new architectures and techniques to extract value.

Why traditional approaches fall short

Aspect	Traditional approach	Big data reality
Scale	Single server or a small cluster; scale **up** (bigger machine) with a hard ceiling	Terabytes to petabytes; needs scale **out** across hundreds of commodity nodes
Data types	Structured rows and columns with a fixed schema	Text, logs, images, video, sensor streams with no fixed schema
Speed	Batch loads, periodic reports	Continuous streams that need near real-time analysis
Storage cost	Expensive SAN and enterprise storage	Cheap distributed storage on commodity hardware
Processing	Move data to the program; one machine's CPU and memory is the bottleneck	Move computation to the data; process in parallel
Schema	Schema-on-write (design first, load later)	Schema-on-read (store raw, interpret later)
Fault tolerance	A server failure can halt processing	Failures are normal at scale, so replication and automatic recovery are required
Analytics	SQL queries and descriptive reports	Machine learning, graph analysis, text and stream analytics

In short, relational databases and single-node tools hit limits in storage capacity, processing speed, data variety and cost, so distributed frameworks such as Hadoop, Spark and NoSQL databases are used instead.

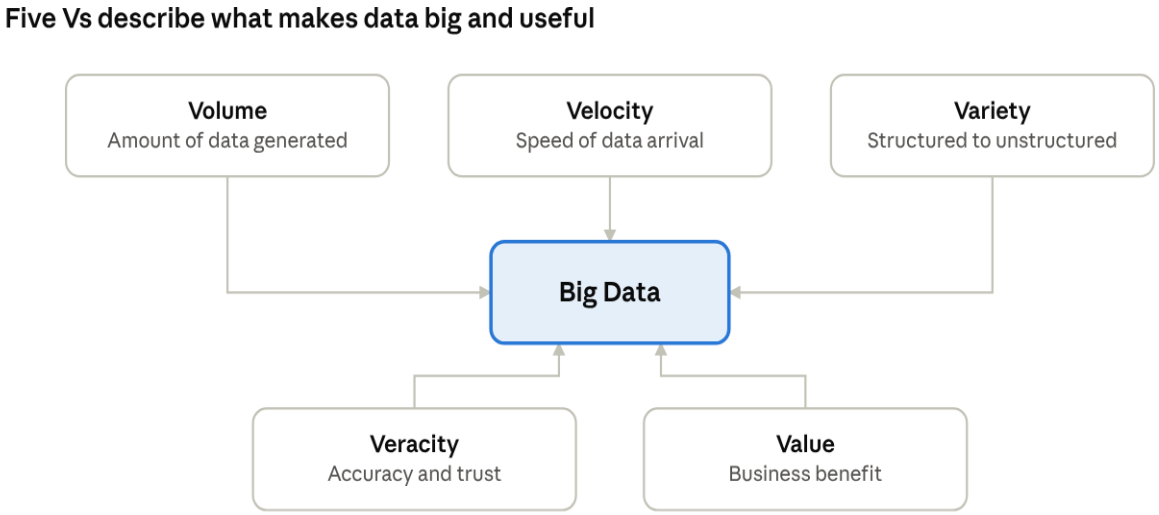
Q2. The 5Vs of Big Data, with practical examples

The 5Vs

1. **Volume:** the sheer amount of data generated, measured in terabytes, petabytes or beyond.
2. **Velocity:** the speed at which data is generated, arrives and must be processed (batch, near real-time or streaming).
3. **Variety:** the many forms of data: structured, semi-structured and unstructured, from many sources.
4. **Veracity:** the quality, accuracy and trustworthiness of data (noise, bias, missing values, duplicates).
5. **Value:** the useful insight and business benefit that can be extracted; data without value is just cost.

Figure

Figure 1 summarises the five Vs around Big Data.



the 5Vs of Big Data · 5 attributes

Volume, velocity and variety describe the data itself; veracity and value decide whether it can be trusted and used.

Practical example: Volume

A social media platform receives billions of photos, posts and messages every day. Storing and analysing this much content requires distributed storage across thousands of servers, because no single machine can hold or process it.

Practical example: Velocity

A credit card company must approve or block a transaction within a fraction of a second. The stream of transactions is analysed in real time against fraud patterns, so waiting for an overnight batch job would let fraud through.

Other quick examples

- **Variety:** a hospital combines patient records (structured), doctor notes (unstructured) and device readings (semi-structured).
- **Veracity:** social media sentiment analysis must filter out bots, sarcasm and fake reviews.
- **Value:** an e-commerce site turns browsing and purchase history into product recommendations that raise sales.

Q3. Structured, semi-structured and unstructured data

Basis	Structured	Semi-structured	Unstructured
Definition	Organised in a fixed schema of rows and columns	Has tags or markers that give some structure, but no rigid schema	No predefined model or organisation
Schema	Fixed, defined before storage	Flexible, self-describing	None
Storage	Relational databases (MySQL, Oracle, PostgreSQL), data warehouses	NoSQL / document stores (MongoDB), files in HDFS	Object storage, data lakes, HDFS, file systems
Querying	Easy with SQL	Possible with query tools or parsers (JSONPath, XPath)	Needs text mining, image or speech processing, machine learning
Examples	Bank transactions, student records table, inventory sheet	JSON, XML, CSV with irregular fields, emails (headers plus body), web server logs, sensor feeds	Images, videos, audio, PDFs, social media posts, free-text documents
Share of data	Small portion (roughly 10 to 20%)	Growing portion	Largest portion (commonly estimated at 80% or more)

Key point: structured data is easiest to analyse, but most of the value in big data comes from combining all three types.

Q4. Role of distributed file systems in big data processing

What they are

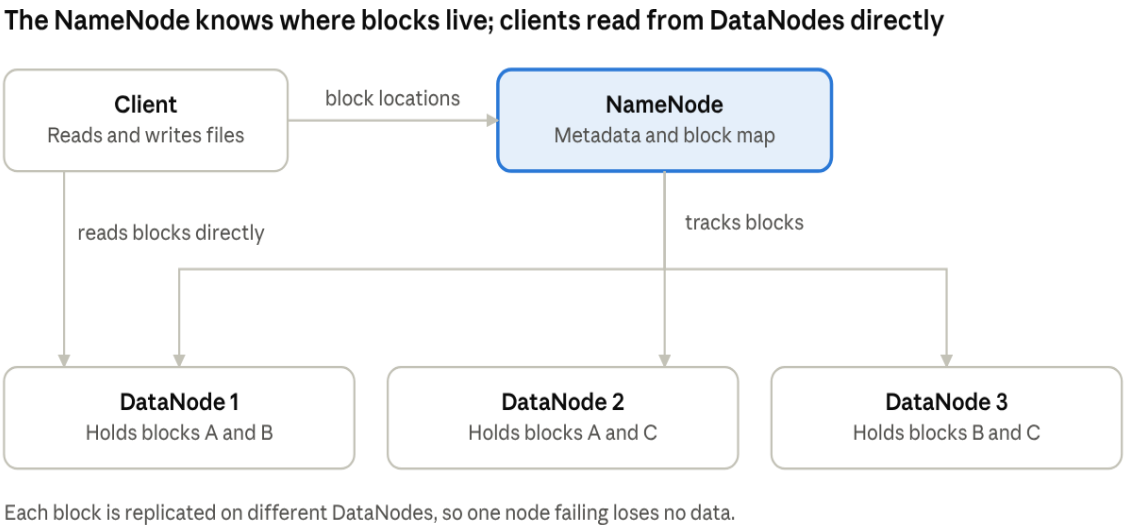
A distributed file system (DFS) stores very large files by splitting them into blocks and spreading those blocks across many machines in a cluster, while presenting one unified file system to users. Examples: Hadoop Distributed File System (HDFS), Google File System (GFS), Amazon S3-based storage layers, Ceph and GlusterFS.

How HDFS works

- A **NameNode** (master) keeps the metadata: file names, block locations and permissions.
- Many **DataNodes** (workers) store the actual blocks, typically 128 MB each.
- Each block is **replicated** (default three copies) on different nodes and racks.
- Clients ask the NameNode where blocks are, then read or write directly with DataNodes.

Figure

Figure 2 shows the HDFS read and replication layout.



HDFS layout · client, NameNode, 3 DataNodes

The client asks the NameNode where blocks are, then reads them straight from the DataNodes that hold copies.

Roles in big data processing

Role	Explanation
Scalable storage	Capacity grows by adding commodity nodes, giving petabyte-scale storage

Role	Explanation
Fault tolerance	Block replication and automatic re-replication keep data available after disk or node failures
Parallel processing	Blocks on many nodes let MapReduce or Spark tasks read and process data simultaneously
Data locality	Computation is sent to the node that holds the data, reducing network traffic
High throughput	Optimised for large sequential reads and writes rather than small random access
Cost efficiency	Runs on cheap commodity hardware instead of expensive storage arrays
Single namespace	Users see one logical file system regardless of where blocks physically live
Foundation for the ecosystem	Hive, HBase, Pig, Spark and other tools use it as their storage layer

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Q5. Three major security concerns in big data platforms

1. **Unauthorized access and weak authentication/authorization.** Big data clusters hold sensitive data accessed by many users and tools. Default configurations in early Hadoop-style platforms had weak authentication, so attackers or over-privileged insiders could read or modify data. Mitigation: strong authentication (Kerberos, multi-factor), role-based access control, fine-grained permissions and regular access audits.
2. **Data privacy and confidentiality breaches.** Large collections of personal, financial or health data attract attackers, and combining datasets can re-identify individuals even after names are removed. Data in transit and at rest may be exposed. Mitigation: encryption in transit and at rest, masking, anonymization or pseudonymization, data classification, and compliance with laws such as GDPR.
3. **Insecure distributed architecture and data in motion.** Many nodes, ports, services and third-party tools expand the attack surface, and one compromised node can expose others; unmonitored data flows between nodes and applications are easy to intercept or tamper with. Mitigation: network segmentation, secure inter-node communication (TLS), patching, hardened configurations, and continuous monitoring and logging.

Other concerns include insider threats, lack of audit trails, insecure third-party tools and data provenance issues.

Q6. Data appliances and integration tools in big data workflows

Data appliances

A data appliance is a pre-integrated package of hardware and software (servers, storage, networking and optimised database or analytics software) built to store and analyse large data volumes with little setup. Examples: Oracle Exadata, IBM Netezza, Teradata appliances, SAP HANA appliances, and Hadoop appliances.

Role: fast deployment, high query performance through tuned hardware and software, simplified management, and predictable scaling for data warehousing and analytics workloads.

Data integration tools

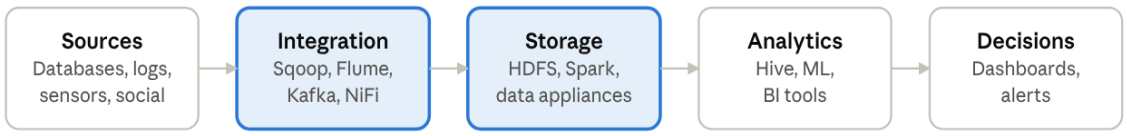
Integration tools collect data from many sources, clean and transform it, and load it into a target system. They handle extract, transform, load (ETL) or extract, load, transform (ELT), data ingestion and streaming. Examples: Apache Sqoop (database to Hadoop), Apache Flume and Kafka (logs and streams), Apache NiFi, Talend, Informatica, Pentaho and AWS Glue.

Role: unify data from databases, applications, logs, sensors and social media; improve quality (deduplication, validation, format conversion); schedule and automate pipelines; and keep data consistent across systems.

Figure

Figure 3 places both within an end-to-end big data workflow.

Integration tools and appliances sit between raw sources and analytics



Highlighted stages are where integration tools and data appliances do their work.

big data workflow · 5 stages

Integration tools move and clean data on the way in, and appliances speed up storage and queries before analytics.

Role in the workflow

Stage	What happens	Typical tools
Sources	Databases, logs, sensors, social media, applications	Operational systems

Stage	What happens	Typical tools
Integration	Ingest, clean, transform, and move data	Sqoop, Flume, Kafka, NiFi, Talend
Storage and processing	Distributed storage and parallel processing; appliances accelerate warehouse-style queries	HDFS, Spark, MapReduce, data appliances
Analytics	Reporting, machine learning, visualization	Hive, Spark MLlib, BI tools
Decisions	Insights drive business action	Dashboards, alerts

Q7. Two major challenges in implementing big data analytics

1. **Data quality, integration and governance.** Data arrives from many systems in different formats, with duplicates, missing values and inconsistent definitions. Siloed data is hard to combine, and without governance (ownership, lineage, privacy rules, standards) results become unreliable and non-compliant. Organisations must invest in cleansing, metadata management, master data practices and clear policies.
2. **Skills shortage and infrastructure and cost complexity.** Big data projects need data engineers, data scientists and analysts who understand distributed systems, statistics and the business domain, and such talent is scarce. Setting up and tuning clusters or cloud platforms, integrating them with legacy systems and controlling storage and compute costs add complexity and risk of failed projects. Remedies include training, managed cloud services, phased pilots and clear business cases.

Other challenges include security and privacy, scalability, organisational resistance and unclear business objectives.

Q8. Short note on a domain application: Healthcare

Healthcare generates huge volumes of electronic health records, medical images, lab results, genomic data, wearable-device readings and insurance claims. Big data analytics turns these into better outcomes and lower costs.

Key uses

- **Predictive analytics:** identifying patients at risk of readmission, diabetes or sepsis so doctors can intervene early.
- **Personalised medicine:** using genomic and history data to choose treatments that suit the individual.
- **Medical imaging analysis:** machine learning models assist in detecting tumours or fractures in scans.

- **Remote monitoring:** wearable and IoT data stream to dashboards that alert clinicians to abnormal readings.
- **Operations:** forecasting hospital admissions, optimising staffing and bed use, and detecting insurance fraud.
- **Public health:** tracking disease outbreaks from clinical and population data.

Benefits: earlier diagnosis, fewer errors, lower cost and improved patient care.

Challenges: privacy and security of patient data, integration of varied record formats and data quality.

(Other valid domains: retail recommendation, banking fraud detection, telecom churn prediction, smart cities, agriculture.)

Q9. How business drivers influence adoption of big data solutions

Business drivers are the strategic needs and pressures that justify investment. They shape what organisations adopt, how fast and how much they spend.

Business driver	Influence on adoption
Competitive advantage	Firms adopt analytics to understand customers and markets faster than rivals
Customer experience and personalisation	Demand for tailored offers and recommendations pushes investment in behavioural analytics
Cost reduction and efficiency	Predictive maintenance, supply chain optimisation and cheap commodity storage justify the spend
Revenue growth and new products	Data-driven products and monetisation of data open new revenue streams
Risk management and compliance	Fraud detection, security analytics and regulatory reporting need large-scale data analysis
Real-time decision making	Need for instant insight drives streaming and in-memory platforms
Data explosion	Growth of IoT, social media and mobile data makes traditional systems inadequate
Technology availability	Open-source tools and cloud services lower entry cost and barriers

Conclusion: adoption succeeds when a project is tied to a measurable business objective (for example, reducing churn by a target percentage), has executive sponsorship, and shows return on investment, rather than being technology-led.

Q10. Importance of big data analytics in decision making

Big data analytics lets organisations replace intuition with evidence by analysing large and varied data quickly.

Types of analytics supporting decisions

1. **Descriptive:** what happened (sales reports, dashboards).
2. **Diagnostic:** why it happened (root-cause analysis).
3. **Predictive:** what is likely to happen (demand forecasting, churn prediction).
4. **Prescriptive:** what should be done (optimised pricing, recommended actions).

Importance

- **Faster, real-time decisions:** streaming analytics lets firms react within seconds, for example blocking fraud or rerouting deliveries.
- **Greater accuracy:** decisions rest on complete data instead of small samples or opinion.
- **Customer insight:** understanding preferences improves targeting, retention and service.
- **Risk reduction:** early warning of fraud, equipment failure or market shifts.
- **Operational efficiency:** identifies waste and optimises supply chains, staffing and inventory.
- **Innovation:** reveals unmet needs and new product or market opportunities.
- **Strategic planning:** trend and scenario analysis supports long-term choices.

Example: a retailer analyses sales, weather and social media trends to forecast demand, stocking the right products in the right stores and cutting stock-outs and waste.
